FIRST LECTURE COURSE HANDOUT

1. Course & Institutional Context

School	School of Computer Science and Engineering
Department	Department of Computer Science and Applications
Program & Specialization	BCA
Academic Year	2026-27
Semester	V
Course Title	Data Mining & Warehousing
Course Code	BCA30120
Course Category	Program Elective - I
Credits & Contact Hours	4 ,45

2. Instructor & Academic Support Details

Course Instructor(s)	Mr. Aamir Mir
Email & Office Hours	Aamir.mir@mitwpu.edu.in
Office Location / Online Link	VY-612/CUB-37
Teaching Assistants	

3. Course Purpose & Relevance Course Rationale

The **Data Mining and Data Warehousing** course provides students with the fundamental concepts, techniques, and tools required to extract meaningful knowledge from large volumes of data and to manage data efficiently for decision-making. The course covers the principles of data warehousing, data preprocessing, association rule mining, classification, clustering, and the practical application of these techniques using the Weka tool.

In today's data-driven world, organizations across domains such as healthcare, banking, retail, ecommerce, finance, telecommunications, manufacturing, and government rely on data analytics to make informed business decisions. This course equips students with the ability to analyze large datasets, discover hidden patterns, build predictive models, and support strategic decisionmaking.

The knowledge gained in this course forms a strong foundation for advanced subjects such as **Machine Learning, Artificial Intelligence, Business Intelligence, Big Data Analytics, Data Science, and Predictive Analytics**. It also develops practical analytical and problem-solving skills that are highly valued in the IT industry.

From a professional perspective, this course prepares students for careers such as **Data Analyst, Business Intelligence Analyst, Data Engineer, Machine Learning Engineer, Data Scientist,**

Database Administrator, and Analytics Consultant. The hands-on exposure to the Weka data mining tool enables students to apply theoretical concepts to real-world datasets, enhancing their industry readiness and employability.

Overall, the course bridges the gap between theoretical concepts and practical applications, enabling students to transform raw data into actionable knowledge for solving real-world business and research problems.

5. Course Outcomes (COs)

CO No.	Course Outcome Statement	Bloom's Level
CO1	Understand the functionality of the various data mining and data warehousing components.	L3
CO2	Choose and apply appropriate data preprocessing techniques.	L3
CO3	Choose the appropriate data mining techniques for real-day analysis problems.	L4
CO4	Use Weka tool to preprocess data and apply data mining techniques to large datasets to extract actionable knowledge.	L3

CO–PO / PSO Mapping (Indicative)

COs →	PO1	PO2	PO3	PO4	PO5
CO1	M	H	L	L	
CO2	M	H	L		
CO3	M	H	L		
CO4	M	H	H	H	

5. Learning Outcomes (As per Course Nature aimed towards granular and measurable knowledge and skills components)

Upon successful completion of this course, students will be able to:

LO1. Explain the concepts, applications, and challenges of data mining and the Knowledge Discovery in Databases (KDD) process.

LO2. Differentiate between operational databases and data warehouses, and describe data warehouse architecture, ETL process, data cubes, and OLAP.

LO3. Apply data preprocessing techniques such as data cleaning, integration, reduction, transformation, and discretization to improve data quality.

LO4. Generate frequent itemsets and association rules using the Apriori algorithm and evaluate their usefulness.

LO5. Apply classification techniques such as Decision Trees and Naïve Bayes to solve predictive data mining problems.

LO6. Perform clustering using K-Means and Hierarchical Clustering algorithms and interpret the clustering results.

LO7. Use the Weka tool to preprocess datasets, implement data mining algorithms, and analyze the generated results for decision-making.

6. Course Structure & Learning Plan

Unit	Major Topics
Unit I	**Introduction to Data Mining** – Data Mining Concepts, KDD Process, Data Types, Pattern Discovery, Data Mining Techniques, Issues and Applications
Unit II	**Introduction to Data Warehousing** – Data Warehouse Concepts, Architecture, ETL Process, Data Warehouse Models, Data Cubes, OLAP, Design and Usage
Unit III	**Data Preprocessing** – Data Cleaning, Data Integration, Data Reduction, Data Transformation, Data Discretization
Unit IV	**Association Rule Mining** – Frequent Itemsets, Association Rules, Apriori Algorithm, Candidate Generation, Rule Generation, Improving Apriori Efficiency
Unit V	**Classification Techniques** – Decision Tree Induction, Attribute Selection Measures, Tree Pruning, Bayes Theorem, Naïve Bayes Classifier, Bayesian Networks
Unit VI	**Clustering Techniques** – Cluster Analysis, Clustering Methods, K-Means Clustering, Hierarchical Clustering
Unit VII	**Introduction to Weka Tool** – Installation, Data Loading, Data Preprocessing, File Formats, Implementation of Association Rule Mining, Classification, and Clustering Algorithms

7. Teaching–Learning Pedagogy

Lectures, Tutorials, Hands-on Laboratory Demonstrations using Weka Tool, Case Studies, Problem-Based Learning (PBL), Experiential Learning through real-world datasets, Collaborative Learning (Group Discussions), Assignments, Self-Learning, and Seminar/Presentation (where applicable).

8. Learning Resources

Type	Details
Textbooks	**1) Data Mining: Concepts and Techniques**, Jiawei Han, Micheline Kamber, Jian Pei, 3rd Edition, Morgan Kaufmann (Elsevier), 2011.
Reference Books	1) **Data Mining and Data Warehousing: Principles and Practical Techniques**, Parteek Bhatia, Cambridge University Press, 1st Edition, 2019. 2) **Advanced Data Mining Techniques**, David L. Olson and Dursun Delen, Springer, 1st Edition, 2008.
Digital / OER Resources (MANDATORY UTILIZATION OF E-RESOURCES in FORMATIVE AND SUMMATIVE ASSESSMENTS AS PER	These e-Resources can be accessed through 1) Weka Documentation & Tutorials – https://www.cs.waikato.ac.nz/ml/weka/ 2) Weka Tutorial – https://www.tutorialspoint.com/weka/index.htm

PROGRAM REQUIREMENTS)	

9. Learning Beyond the Syllabus

Advanced topics, interdisciplinary exposure, research papers, certifications as applicable.

Advanced Topics

- Big Data Analytics using Hadoop and Spark
- Machine Learning for Data Mining
- Deep Learning and Neural Networks
- Web and Text Mining
- Social Network Analysis
- Cloud-based Data Warehousing
- Data Lakes and Lakehouses
- Real-time Data Analytics and Stream Mining
- Explainable AI (XAI) and Responsible AI

Interdisciplinary Exposure

- Healthcare Analytics
- Financial Fraud Detection
- Retail and Market Basket Analysis
- Cybersecurity Analytics
- Smart Cities and IoT Data Analytics
- Business Intelligence and Decision Support Systems

Research Papers

Students are encouraged to read recent research articles from:

- IEEE Xplore Digital Library
- ACM Digital Library
- SpringerLink
- ScienceDirect
- Google Scholar

Industry Certifications (Optional)

- Oracle Database SQL Certified Associate
- Microsoft Azure Data Fundamentals (DP-900)
- Microsoft Power BI Data Analyst (PL-300)
- Google Data Analytics Professional Certificate
- IBM Data Science Professional Certificate
- AWS Certified Data Engineer – Associate (or AWS Data Analytics learning path)

Self-Learning Activities

- Implement data mining algorithms on real-world datasets from Kaggle and the UCI Machine Learning Repository.
- Explore Weka, Orange, RapidMiner, or KNIME for comparative analysis.

- Participate in data analytics hackathons, coding competitions, and Kaggle challenges.
- Study recent trends in AI-driven data mining and business intelligence. **10. Formative and Summative Assessment & Evaluation Framework**

REFER TO THE TABLE BELOW:

Sr. No.	Assessment Component	Planned Week	Weightage (%)
1	CCA1/ LCA1 Sub Component 1 **(Formative)**	Week 3	ZERO WEIGHTAGE BUT FORMATIVE MODE . Marks/ grades given to be used for Students Academic Standing and Counselling. Don't Count it for CCA1 final marks.
2	CCA1 /LCA 1 Sub Component 2 **(Formative)**	Week 5	ZERO WEIGHTAGE BUT FORMATIVE MODE . Marks/ grades given to be used for Students Academic
			Standing and Counselling. Don't Count it for CCA1 final marks.
3	CCA1 /LCA 1 **(Summative)**	Week 6	Weightage As per Assessment Code. **Marks given to be counted and entered in ERP as per dates given in Academic Calendar**
4	Mid Term (Formative/ Summative as per Schools Format)	Week 10	Weightage As per Assessment Code **. Marks given to be counted and entered in ERP as per dates given in Academic Calendar**
5	**CCA/LCA Post Mid term to have Sub Component in formative mode where marks would not be counted and not to be entered in ERP as given above CCA1/LCA1 subcomponent**	Week 12	ZERO WEIGHTAGE BUT FORMATIVE MODE **. Marks/ grades given to be used for Students Academic Standing and Counselling. Don't Count it for post mid term CCA final marks.**
6	CCA/LCA Final Component in Summative Mode.	Week 15	Weightage As per Assessment Code **. Marks given to be counted and entered in ERP as per dates given in Academic Calendar**
7	End Term Summative- As per Assessment Code	Week 17	**As per Examination Notifications and Schedule**

10. Attendance, Participation & Academic Conduct

As per University Academic Regulations Minimum 80% Attendance Mandatory. BCI,PCI, CoA programs to have their own attendance criteria as per their statutory requirement.

11. Student Responsibilities

Preparation, participation, integrity, timely submission.

12. Feedback & Academic Support

Formative feedback, mentoring, counselling support.

13. Declaration

This document outlines academic expectations for the course.

Prepared by:

Course Faculty Name-Mr.Rajesh Kanzade

Date: 06 July 2026